

Kianoosh Vadaei

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PERSONAL PROFILE

I am a B.S. computer engineering student at the University of Isfahan with a strong interest in AI and machine learning research, particularly in multimodal and self-supervised representation learning. I am motivated by developing learning systems that are robust, generalizable, and applicable beyond controlled settings.

EDUCATION

University of Isfahan

Sept 2021 – Present

B.S. Computer Engineering

- **GPA:** 3.6/4.0 (17.35/20.0)
- **GPA (last four semesters):** 3.77/4.0
- **Thesis topic:** A Unified Self-Supervised Learning Framework Across Modalities

RELEVANT COURSES

Fundamentals of Machine Learning (20/20) – Fundamentals of Data Mining (20/20) – Fundamentals of IoT (20/20) – Applications of Artificial Intelligence in Cybersecurity (20/20) – Special Topics - Deep Learning (19.8/20) – Fundamentals of Computational Intelligence (19/20) – Fundamentals of Artificial Intelligence (17.75/20).

RESEARCH INTERESTS

Multimodal and Self-Supervised Representation Learning | 3D Visual Understanding | Molecular Graph Representation Learning | Explainable and Interpretable AI | AI for Science

PUBLICATIONS

Adaptive Chunking for VideoRAG Pipelines with a Newly Gathered Bilingual Educational Dataset CSICC 2025

Arshia Hemmat, **Kianoosh Vadaei**, Melika Shirian, Mohammad Hassan Heydari, Afsaneh Fatemi
Published (Oral). [\[Link\]](#)

From Scenes to Semantics: PersianCLEVR for Bilingual 3D Visual Reasoning NeurIPS 2025

Kianoosh Vadaei, Melika Shirian, Arshia Hemmat, Mohammad Hassan Heydari, Ali Mamanpoosh, Afsaneh Fatemi
Published (Spotlight) — VLM4RWD Workshop. [\[Link\]](#)

Leveraging Retrieval-Augmented Generation for Persian University Knowledge Retrieval IKT 2024

Arshia Hemmat, **Kianoosh Vadaei**, Mohammad Hassan Heydari, Afsaneh Fatemi
Published (Oral). [\[Link\]](#)

PrismSSL: One Interface, Many Modalities; A Single-Interface Library for Multimodal Self-Supervised Learning Under review (EACL 2026)

Melika Shirian, **Kianoosh Vadaei**, Kian Majlessi, Audrina Ebrahimi, Arshia Hemmat, Peyman Adibi, Hossein Karshenas
Preprint (arXiv:2511.17776). [\[Link\]](#)

Numbers Are Not Easy: A Benchmark for Robust Numerical Reasoning Under review (ACL 2026)

Authors withheld for anonymous review

RESEARCH EXPERIENCE

A Unified Self-Supervised Learning Framework | *ILS Lab - University of Isfahan* May 2025 – September 2025

- Conducted my Bachelor's thesis on self-supervised learning to build an open-source framework for scalable representation learning across multiple data domains.
- I gained hands-on experience with contrastive learning techniques, including zero-shot classification via embedding similarity, CTC-based fine-tuning with WER/PER evaluation, molecular graph representation learning, and linear-probing, implemented from scratch using PyTorch.
- I collaboratively developed and packaged the framework as a reusable Python library, presented the work to an academic panel, and received a final grade of 20/20.

- I explored video information retrieval benchmarks and contributed to the development of adaptive chunking strategies for educational videos, focusing on identifying semantically meaningful temporal segments.
- I experimented with entropy-based and multimodal similarity methods to select informative video chunks, resulting in an approach that was released as an open-source Python package for adaptive VideoRAG pipelines.

PersianCLEVR for Bilingual 3D Visual Reasoning | *University of Isfahan*

February 2025 – October 2025

- I worked on preparing the PersianCLEVR dataset by developing prompt-engineering pipelines for 3D CLEVR-family data translation and structured question-answer generation.
- I ran and analyzed multiple open-source and proprietary vision-language models at scale, evaluating their responses using both statistical metrics and LLM-as-judge frameworks under different prompting settings.
- I presented this work at the NeurIPS 2025 VLM4RWD workshop, gaining experience in clearly communicating technical results to an international academic audience.

Persian University RAG system | *University of Isfahan AI Association*

July 2024 – December 2024

- I worked with LangChain-based RAG pipelines, helped curate and organize university-domain documents, and evaluated system performance using RAGAS metrics.
- Developed a Streamlit-based chatbot to present the research system, accessible for real student use.

Smart Home Assistant | *Hoopad Company*

Summer 2024

- Carried out AI research on Siamese networks for face recognition, leveraging triplet loss to improve accuracy.
- Contributed to real-time face recognition research with a focus on accuracy and embedded system deployment.

ACADEMIC SERVICE**Teaching Assistantships**

- Served as a teaching assistant across 15+ course sections, contributing to project and assignment design, student mentoring, and instructional support.
- Courses assisted: **Fundamentals of Computer Programming** (4 sections), **Advanced Programming** (4 sections), **Algorithms Analysis and Design**, **Computer Architecture** (3 sections), **Fundamentals of IoT**, **Microprocessors and Assembly Language**, **Applications of Artificial Intelligence in Cybersecurity**, **Computer Skills Workshop** (3 sections).

Peer Reviewing

- Reviewed 3 research papers for the NeurIPS 2025 Workshop on Vision-Language Models for Real-World Dynamics (VLM4RWD) via OpenReview.

LEADERSHIP AND AWARDS**Academic & Community Leadership**

- **Chairman, University of Isfahan AI Community (UIAIC):** Elected twice to lead the university AI community, overseeing technical events, workshops, and AI-focused competitions.
- **Editor-in-Chief, *Gradient Magazine*:** Directed editorial strategy and publication processes for UIAIC's official technical magazine.
- **Organizer, TEDx University of Isfahan:** Core organizer of the 4th TEDx event (100+ attendees) and co-organizer of the 5th edition, coordinating multiple teams and ensuring compliance with TEDx guidelines. [Link]
- **Executive Committee Member, 7th International Conference on Internet of Things and Its Applications:** Contributed to conference coordination, communications, and operational planning.
- **Executive Head, Rasta Association Summer School 2024 (Sharif University of Technology):** Led academic and organizational planning for a national-level summer school program.

SKILLS**Technical Languages:** Python (Advanced), C#, C++, L^AT_EX**Frameworks & Libraries:** PyTorch (Advanced), Transformers, Pandas, Scikit-learn, NumPy**Tools:** Git, Weights & Biases (W&B)**Languages:** IELTS 7.5 (Listening: 8.0, Reading: 7.5, Writing: 7.0, Speaking: 7.0), C1